

TidyTuesday Week 39

Pranay Gundam

Sunday 27th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the only weekly column brave enough to pair economic time series like a blindfolded matchmaker and then act surprised when they don't get along. As always, the disclaimer that keeps me out of statistical jail: these are randomly chosen series with zero reason to be related, and any correlation you see here is the universe's idea of a practical joke. Onward.

Monday kicked things off with New England commercial electricity prices versus a price index for medical expenditures on, and I quote, "symptoms, signs, and ill-defined conditions." The raw regression came out screaming, R-squared of 0.93, a p-value with more zeros than my bank account. Absolutely thrilling! Right up until you detrend it, at which point the whole thing evaporates into an R-squared of 0.02 and a p-value of basically "who knows." Classic trend mirage. Two things went up over time, held hands, and pretended it meant something. It did not. Tuesday gave us a rarer treat: Montana housing listings in Flathead County against U.S. imports of maintenance and repair services. Weaker to begin with (raw R-squared of 0.12), but here's the twist – the relationship actually got STRONGER after detrending, up to 0.19 and still very significant. I have no idea why active real estate listings in one Montana county would relate to national repair-service imports, and I refuse to invent a theory, but I'll grudgingly tip my hat: it survived the detrending guillotine, which is more than most contestants manage.

Wednesday was the week's overachiever. Wholesale-trade payrolls versus the Dover, Delaware house price index posted a raw R-squared of 0.73, and – brace yourself – it held onto 0.73 after detrending too, p-value still buried in the scientific-notation basement. Two independent series that both drift AND wiggle together? Suspicious. Probably still coincidence dressed in a nice suit, because payrolls and house prices both quietly track "the economy doing economy things," but as far as this project goes, that's a genuine standout worth pointing at. Thursday, meanwhile, gave us Southwest central-bank GDP against a gloriously named discontinued bank-assets series, where the raw result was borderline (p of 0.05, a coward's significance) but the detrended version firmed up to a real negative relationship. Interpret at your own peril.

Friday paired education-and-communication consumer prices with total tax exemptions under age 65 in New Mexico, and this one actually doubled down: R-squared jumped from 0.32 raw to 0.51 detrended. Two more series conspiring behind the trend's back. Then the weekend arrived to humble us all. Saturday's "shares of gross domestic income" versus a Peabody, Massachusetts house price index produced a detrended coefficient of roughly 57 trillion with a p-value of 0.82 and warnings about a singular matrix – which is math's polite way of saying "please stop." And Sunday's employment cost index versus U.S. JPY intervention transactions returned nothing but

zeros and NaNs, the regression equivalent of the machine unplugging itself and walking out. Fitting end to the week: a strong reminder that spurious is the house specialty, the occasional survivor is just noise with good posture, and I'll be right back here next week to do it all again.

2 Date: 2026-09-21

Series ID: PCU221122221122421

This series is titled Producer Price Index by Industry: Electric Power Distribution: Commercial Electric Power for New England Census Division and has a frequency of Monthly. The units are Index Dec 1990=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1971-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: SYSIILPIBLEND

This series is titled Medical Services Expenditures by Disease: Symptoms; Signs; and Ill-Defined Conditions Price Index, Blended Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SYSIILPIBLEND	R-squared:	0.933
Model:	OLS	Adj. R-squared:	0.929
Method:	Least Squares	F-statistic:	277.7
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	3.41e-13
Time:	17:16:08	Log-Likelihood:	-68.300
No. Observations:	22	AIC:	140.6
Df Residuals:	20	BIC:	142.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.0876	5.281	-1.153	0.263	-17.103	4.928
value_fred_PCU221122221122421	0.5458	0.033	16.664	0.000	0.477	0.614

Omnibus:	6.881	Durbin-Watson:	1.268
Prob(Omnibus):	0.032	Jarque-Bera (JB):	5.911
Skew:	0.534	Prob(JB):	0.0521
Kurtosis:	5.304	Cond. No.	706.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

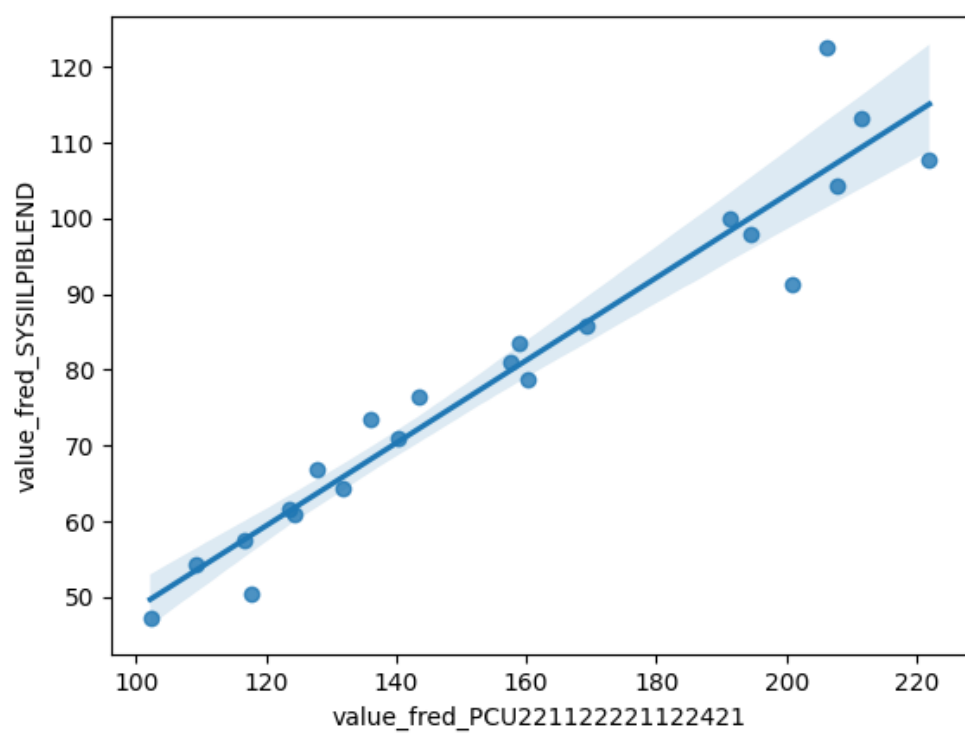


Figure 1: Raw Regression Plot for 2026-09-21

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SYSIILPIBLEND_detrended	R-squared:	0.022
Model:	OLS	Adj. R-squared:	-0.027
Method:	Least Squares	F-statistic:	0.4428
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	0.513
Time:	17:16:08	Log-Likelihood:	-54.441
No. Observations:	22	AIC:	112.9
Df Residuals:	20	BIC:	115.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.119e-14	0.643	6.41e-14	1.000	-1.340	1.340
value_fred_PCU221122221122421_detrended	0.0480	0.072	0.665	0.513	-0.103	0.199

Omnibus:	12.738	Durbin-Watson:	0.457
Prob(Omnibus):	0.002	Jarque-Bera (JB):	12.678
Skew:	1.200	Prob(JB):	0.00177
Kurtosis:	5.841	Cond. No.	8.90

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

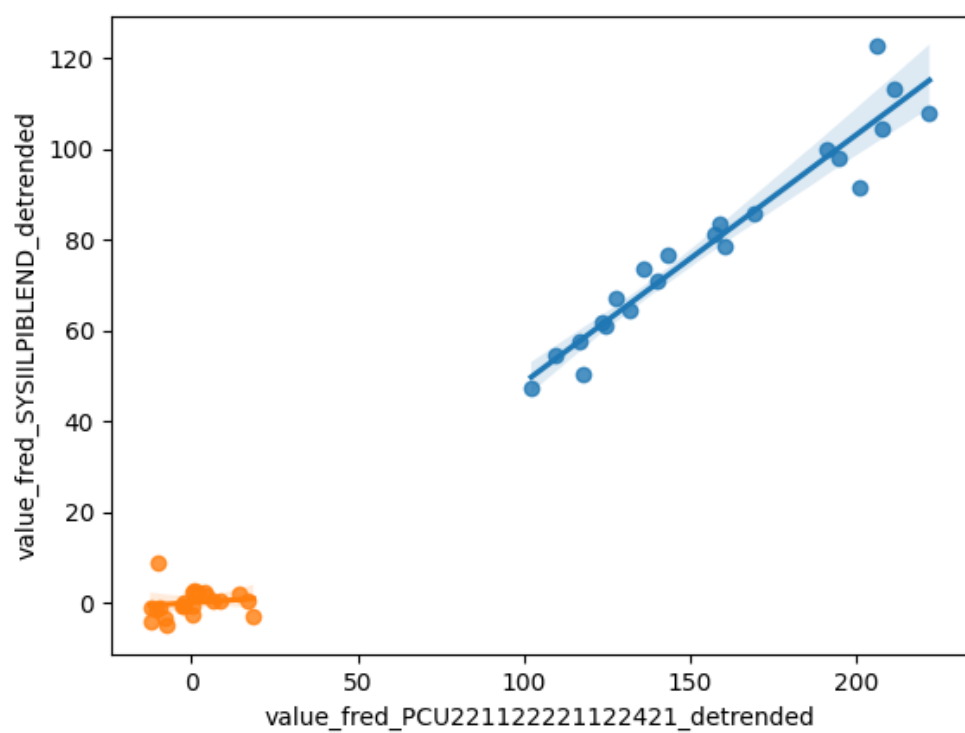


Figure 2: Detrended Regression Plot for 2026-09-21

3 Date: 2026-09-22

Series ID: ACTLISCOU30029

This series is titled Housing Inventory: Active Listing Count in Flathead County, MT and has a frequency of Monthly. The units are Level and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2016-07-01 and the observation end date is 2026-08-01. The popularity of this series is 2.

Series ID: ITMMARM133S

This series is titled U.S. Imports of Services: Maintenance and Repair Services, not included elsewhere and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITMMARM133S	R-squared:	0.123
Model:	OLS	Adj. R-squared:	0.116
Method:	Least Squares	F-statistic:	16.76
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	7.76e-05
Time:	15:49:46	Log-Likelihood:	-747.35
No. Observations:	121	AIC:	1499.
Df Residuals:	119	BIC:	1504.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	452.9830	36.286	12.484	0.000	381.134	524.832
value_fred_ACTLISCOU30029	0.1837	0.045	4.094	0.000	0.095	0.273

Omnibus:	12.594	Durbin-Watson:	0.214
Prob(Omnibus):	0.002	Jarque-Bera (JB):	14.022
Skew:	0.833	Prob(JB):	0.000902
Kurtosis:	3.056	Cond. No.	2.75e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.75e+03. This might indicate that there are strong multicollinearity or other numerical problems.

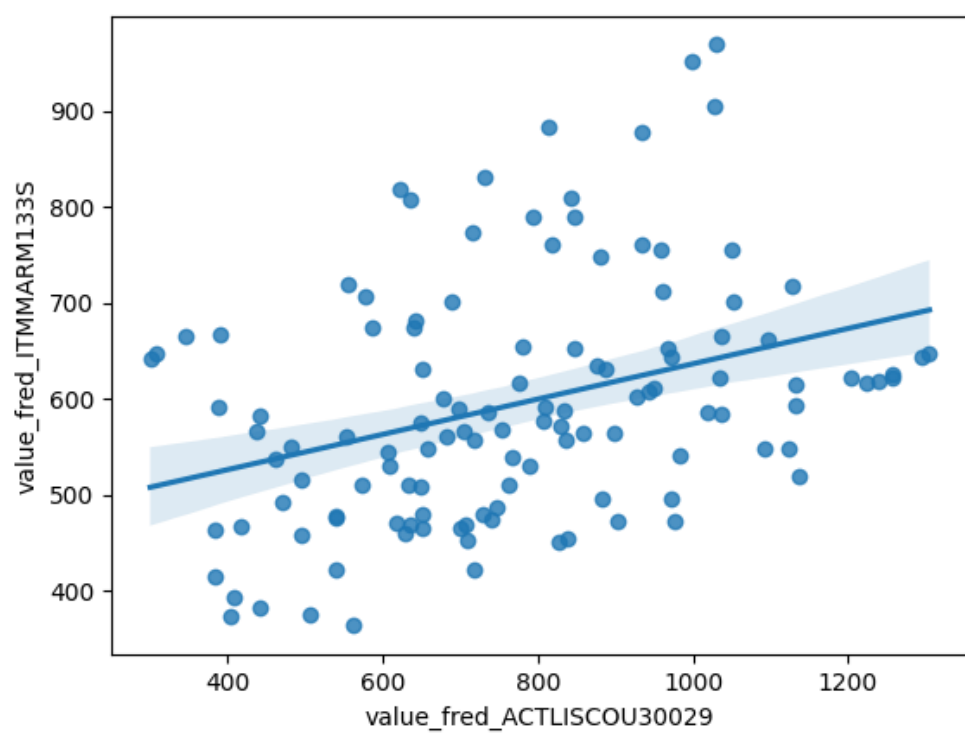


Figure 3: Raw Regression Plot for 2026-09-22

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITMMARM133S_detrended	R-squared:	0.195
Model:	OLS	Adj. R-squared:	0.188
Method:	Least Squares	F-statistic:	28.77
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	4.07e-07
Time:	15:49:46	Log-Likelihood:	-740.70
No. Observations:	121	AIC:	1485.
Df Residuals:	119	BIC:	1491.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-3.037e-11	10.104	-3.01e-12	1.000	-20.008	20.008
value_fred_ACTLISCOU30029_detrended	0.2437	0.045	5.364	0.000	0.154	0.334

Omnibus:	4.932	Durbin-Watson:	0.254
Prob(Omnibus):	0.085	Jarque-Bera (JB):	5.006
Skew:	0.473	Prob(JB):	0.0819
Kurtosis:	2.689	Cond. No.	222.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

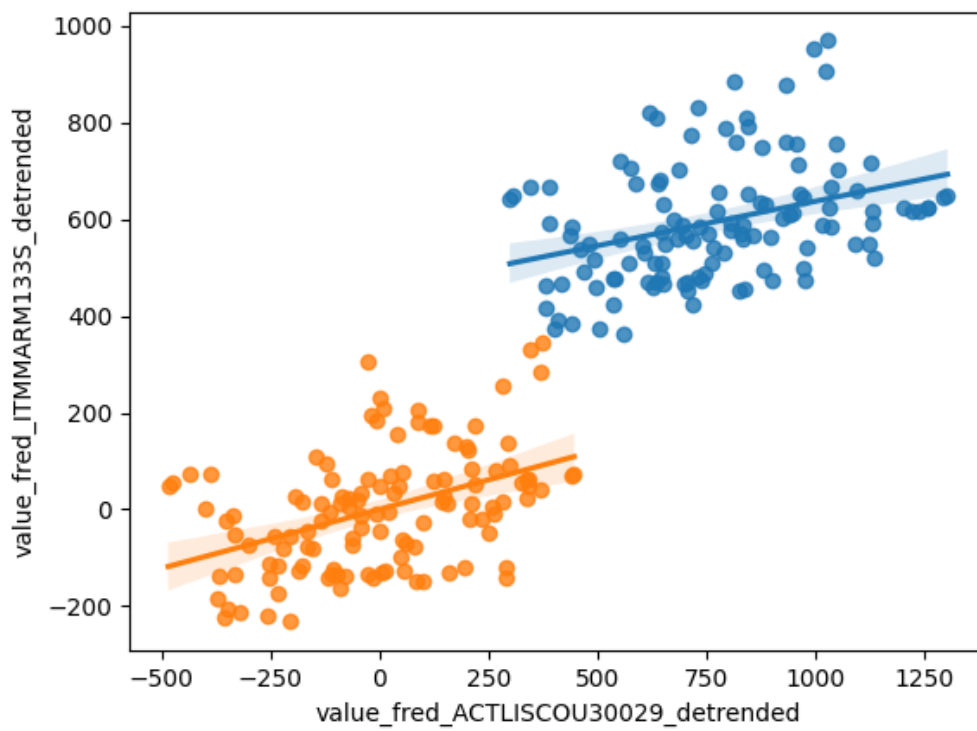


Figure 4: Detrended Regression Plot for 2026-09-22

4 Date: 2026-09-23

Series ID: CEU4142000017

This series is titled Indexes of Aggregate Weekly Payrolls of All Employees, Wholesale Trade and has a frequency of Monthly. The units are Index 2007=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-03-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: ATNHPIUS20100Q

This series is titled All-Transactions House Price Index for Dover, DE (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1987-07-01 and the observation end date is 2026-04-01. The popularity of this series is 2.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ATNHPIUS20100Q	R-squared:	0.731			
Model:	OLS	Adj. R-squared:	0.728			
Method:	Least Squares	F-statistic:	214.8			
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	3.06e-24			
Time:	15:39:04	Log-Likelihood:	-377.92			
No. Observations:	81	AIC:	759.8			
Df Residuals:	79	BIC:	764.6			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-11.5081	15.435	-0.746	0.458	-42.230	19.214
value_fred_CEU4142000017	1.7805	0.121	14.656	0.000	1.539	2.022
Omnibus:	32.589	Durbin-Watson:	0.083			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	6.325			
Skew:	0.272	Prob(JB):	0.0423			
Kurtosis:	1.744	Cond. No.	678.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

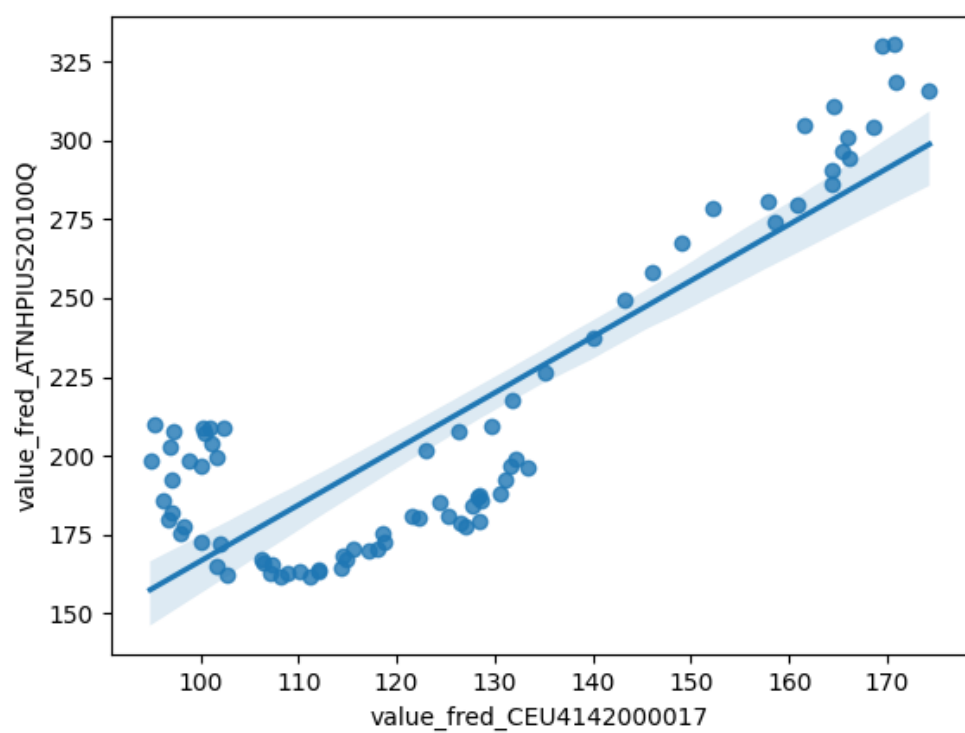


Figure 5: Raw Regression Plot for 2026-09-23

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ATNHPIUS20100Q_detrended	R-squared:	0.727
Model:	OLS	Adj. R-squared:	0.724
Method:	Least Squares	F-statistic:	210.7
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	5.32e-24
Time:	15:39:04	Log-Likelihood:	-349.22
No. Observations:	81	AIC:	702.4
Df Residuals:	79	BIC:	707.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.785e-12	2.029	-3.84e-12	1.000	-4.039	4.039
value_fred_CEU4142000017_detrended	4.4163	0.304	14.517	0.000	3.811	5.022

Omnibus:	7.128	Durbin-Watson:	0.420
Prob(Omnibus):	0.028	Jarque-Bera (JB):	3.489
Skew:	0.261	Prob(JB):	0.175
Kurtosis:	2.128	Cond. No.	6.67

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

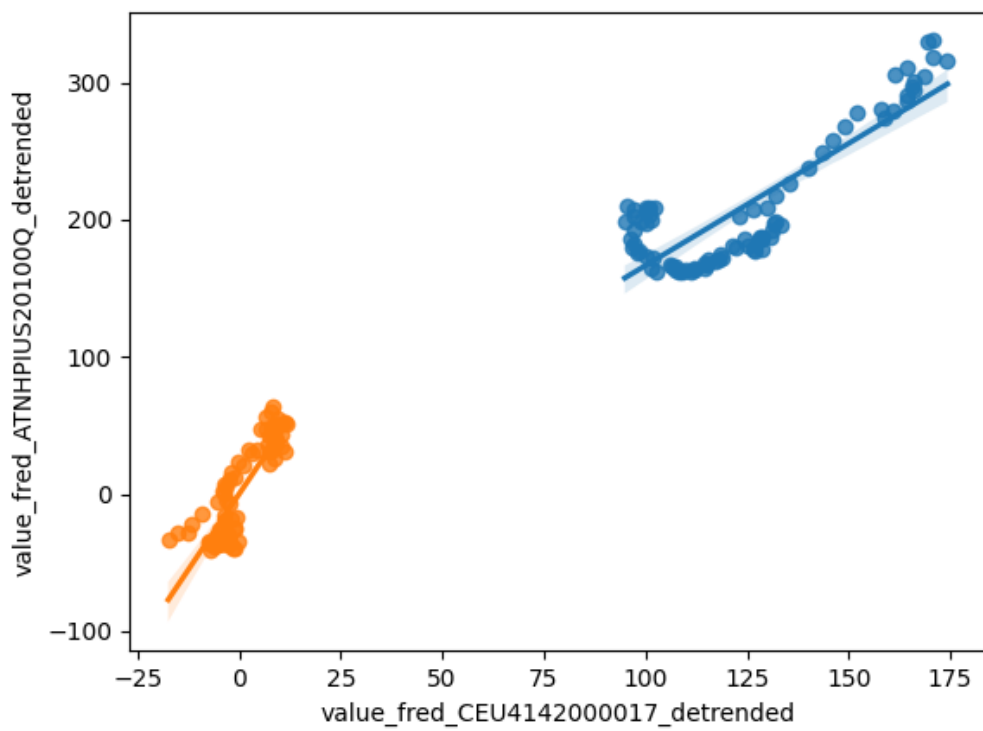


Figure 6: Detrended Regression Plot for 2026-09-23

5 Date: 2026-09-24

Series ID: SWSTFRBCIRGSP

This series is titled Real Gross Domestic Product: Monetary Authorities-Central Bank, Credit Intermediation, and Related Services (521-522) in the Southwest BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: LLRNPT39

This series is titled Assets at Banks whose ALLL exceeds their Nonperforming Loans, Banks with Total Assets from \$1B to \$10B, Pacific Census Division (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 2020-07-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LLRNPT39	R-squared:	0.158
Model:	OLS	Adj. R-squared:	0.119
Method:	Least Squares	F-statistic:	4.120
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.0546
Time:	15:59:56	Log-Likelihood:	-110.56
No. Observations:	24	AIC:	225.1
Df Residuals:	22	BIC:	227.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	152.7995	40.101	3.810	0.001	69.635	235.964
value_fred_SWSTFRBCIRGSP	-0.0016	0.001	-2.030	0.055	-0.003	3.37e-05

Omnibus:	4.915	Durbin-Watson:	0.325
Prob(Omnibus):	0.086	Jarque-Bera (JB):	3.660
Skew:	-0.955	Prob(JB):	0.160
Kurtosis:	3.111	Cond. No.	4.07e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

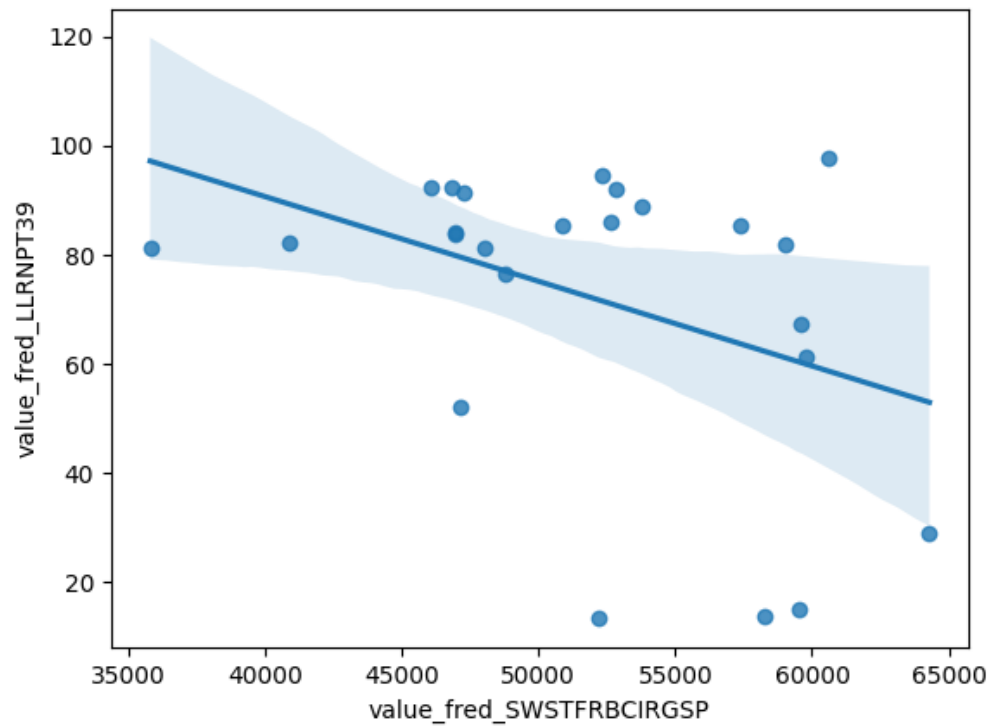


Figure 7: Raw Regression Plot for 2026-09-24

[2] The condition number is large, $4.07e+05$. This might indicate that there are strong multicollinearity or other numerical problems.

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LLNRPT39_detrended	R-squared:	0.277
Model:	OLS	Adj. R-squared:	0.244
Method:	Least Squares	F-statistic:	8.412
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.00830
Time:	15:59:56	Log-Likelihood:	-108.67
No. Observations:	24	AIC:	221.3
Df Residuals:	22	BIC:	223.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.199e-12	4.776	2.51e-13	1.000	-9.904	9.904
value_fred_SWSTFRBCIRGSP_detrended	-0.0031	0.001	-2.900	0.008	-0.005	-0.001

Omnibus:	7.727	Durbin-Watson:	0.515
Prob(Omnibus):	0.021	Jarque-Bera (JB):	5.892
Skew:	-1.175	Prob(JB):	0.0526
Kurtosis:	3.608	Cond. No.	4.43e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

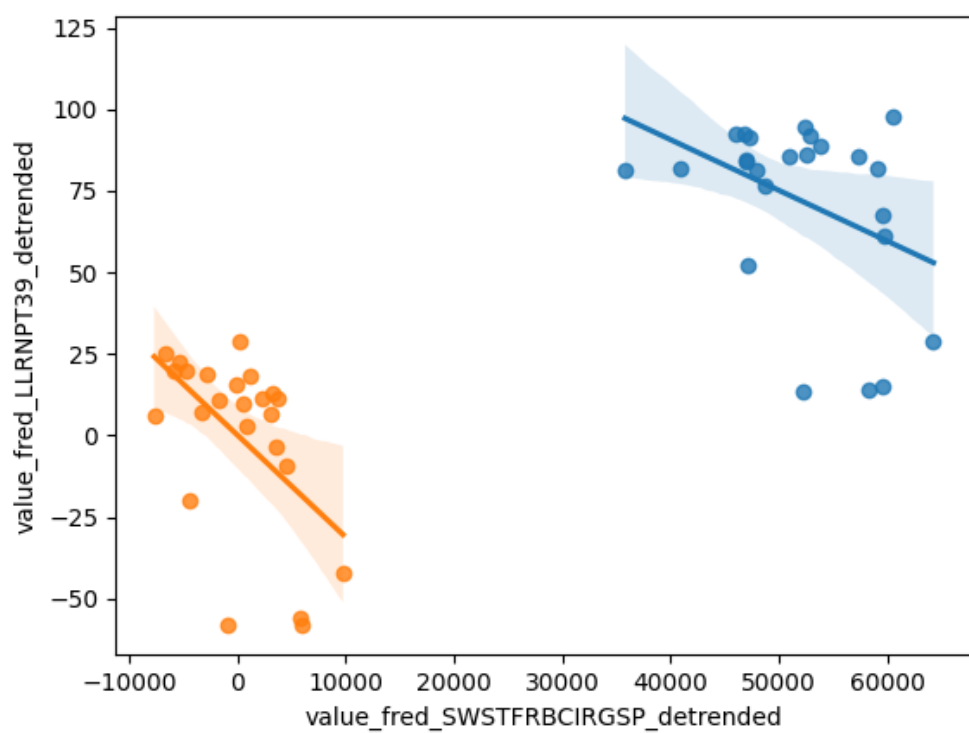


Figure 8: Detrended Regression Plot for 2026-09-24

6 Date: 2026-09-25

Series ID: CWSR0000SAE

This series is titled Consumer Price Index for All Urban Wage Earners and Clerical Workers: Education and Communication in U.S. City Average and has a frequency of Monthly. The units are Index Dec 1997=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1993-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: LT65EXMNM35A647NCEN

This series is titled Total Tax Exemptions Under Age 65 for New Mexico and has a frequency of Annual. The units are Number of Exemptions and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1989-01-01 and the observation end date is 2023-01-01. The popularity of this series is 1.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LT65EXMNM35A647NCEN	R-squared:	0.319
Model:	OLS	Adj. R-squared:	0.296
Method:	Least Squares	F-statistic:	13.61
Date:	Fri, 25 Sep 2026	Prob (F-statistic):	0.000923
Time:	16:02:20	Log-Likelihood:	-383.39
No. Observations:	31	AIC:	770.8
Df Residuals:	29	BIC:	773.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.111e+06	8.56e+04	12.985	0.000	9.36e+05	1.29e+06
value_fred_CWSR0000SAE	2729.6493	739.846	3.689	0.001	1216.494	4242.805

Omnibus:	1.749	Durbin-Watson:	0.439
Prob(Omnibus):	0.417	Jarque-Bera (JB):	1.177
Skew:	0.477	Prob(JB):	0.555
Kurtosis:	2.985	Cond. No.	937.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

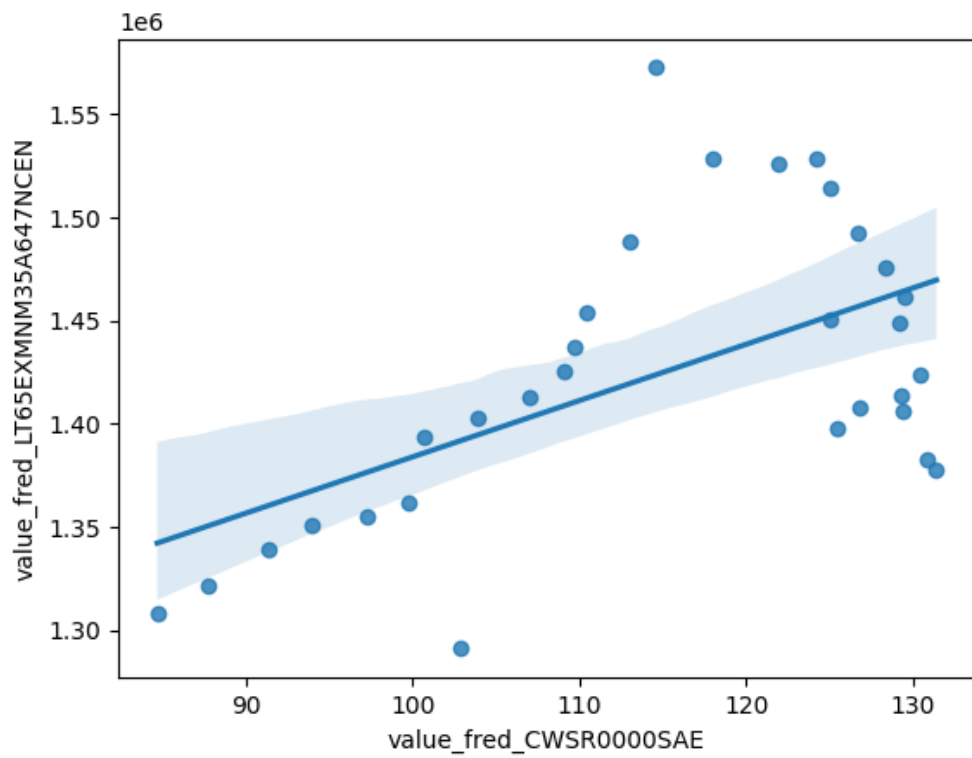


Figure 9: Raw Regression Plot for 2026-09-25

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LT65EXMNM35A647NCEN_detrended	R-squared:	0.508
Model:	OLS	Adj. R-squared:	0.491
Method:	Least Squares	F-statistic:	29.99
Date:	Fri, 25 Sep 2026	Prob (F-statistic):	6.75e-06
Time:	16:02:20	Log-Likelihood:	-375.82
No. Observations:	31	AIC:	755.6
Df Residuals:	29	BIC:	758.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.538e-09	8271.954	5.49e-13	1.000	-1.69e+04	1.69e+04
value_fred_CWSR0000SAE_detrended	1.073e+04	1959.312	5.477	0.000	6722.977	1.47e+04

Omnibus:	6.655	Durbin-Watson:	0.863
Prob(Omnibus):	0.036	Jarque-Bera (JB):	6.187
Skew:	0.543	Prob(JB):	0.0453
Kurtosis:	4.900	Cond. No.	4.22

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

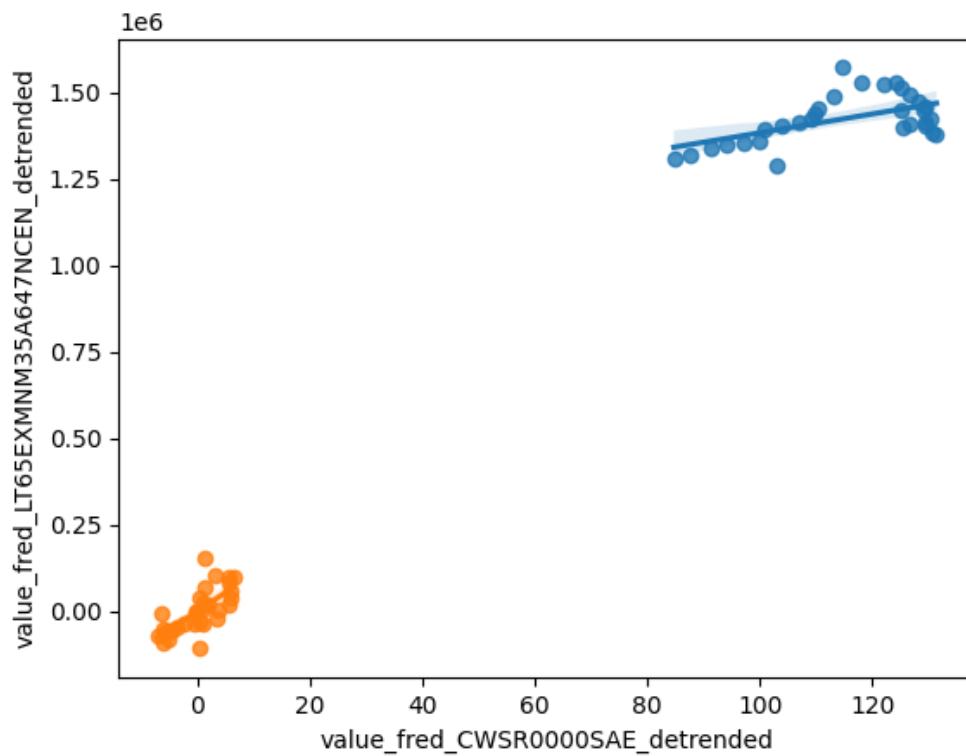


Figure 10: Detrended Regression Plot for 2026-09-25

7 Date: 2026-09-26

Series ID: A261RE1A156NBEA

This series is titled Shares of gross domestic income and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2024-01-01. The popularity of this series is 3.

Series ID: ATNHPIUS37764Q

This series is titled All-Transactions House Price Index for Peabody, MA (MSAD) (DISCONTINUED) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1979-07-01 and the observation end date is 2013-01-01. The popularity of this series is 1.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ATNHPIUS37764Q	R-squared:	-0.000			
Model:	OLS	Adj. R-squared:	-0.000			
Method:	Least Squares	F-statistic:	nan			
Date:	Sat, 26 Sep 2026	Prob (F-statistic):	nan			
Time:	15:12:54	Log-Likelihood:	-191.64			
No. Observations:	34	AIC:	385.3			
Df Residuals:	33	BIC:	386.8			
Df Model:	0					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
value_fred_A261RE1A156NBEA	1.4129	0.118	11.962	0.000	1.173	1.653
	Omnibus:	8.793	Durbin-Watson:	0.027		
	Prob(Omnibus):	0.012	Jarque-Bera (JB):	2.581		
	Skew:	0.231	Prob(JB):	0.275		
	Kurtosis:	1.732	Cond. No.	1.00		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

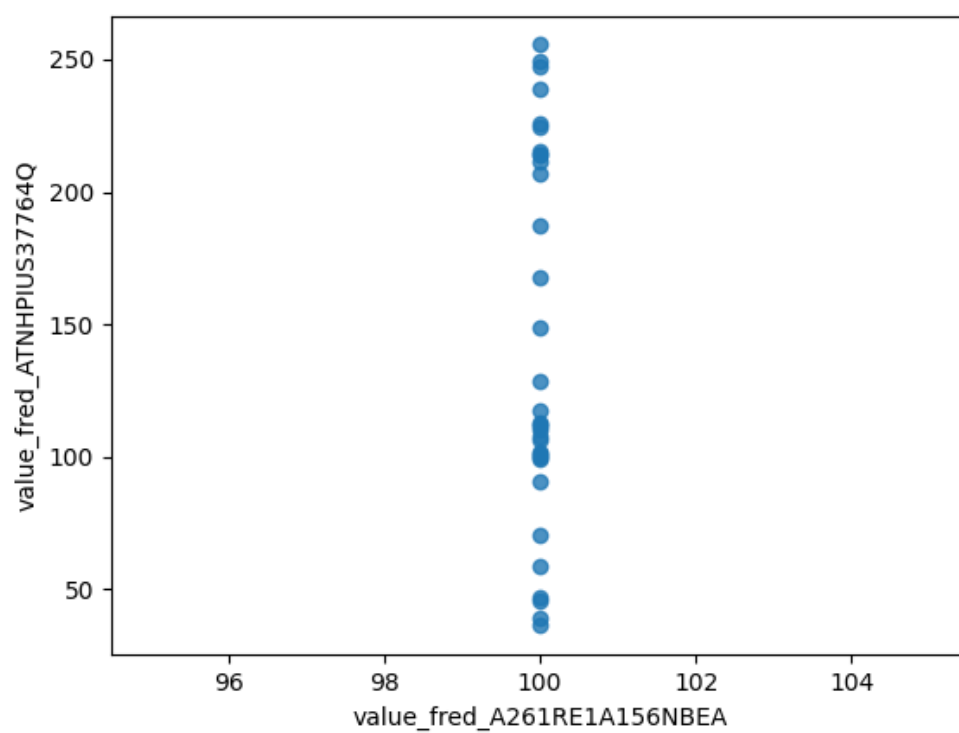


Figure 11: Raw Regression Plot for 2026-09-26

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ATNHPIUS37764Q_detrended	R-squared:	0.002
Model:	OLS	Adj. R-squared:	-0.030
Method:	Least Squares	F-statistic:	0.04974
Date:	Sat, 26 Sep 2026	Prob (F-statistic):	0.825
Time:	15:12:54	Log-Likelihood:	-157.44
No. Observations:	34	AIC:	318.9
Df Residuals:	32	BIC:	321.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-29.7625	133.528	-0.223	0.825	-301.750	242.225
value_fred_A261RE1A156NBEA_detrended	5.71e+13	2.56e+14	0.223	0.825	-4.64e+14	5.79e+14

Omnibus:	2.062	Durbin-Watson:	0.159
Prob(Omnibus):	0.357	Jarque-Bera (JB):	1.548
Skew:	0.330	Prob(JB):	0.461
Kurtosis:	2.190	Cond. No.	5.84e+13

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 9.98e-27. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

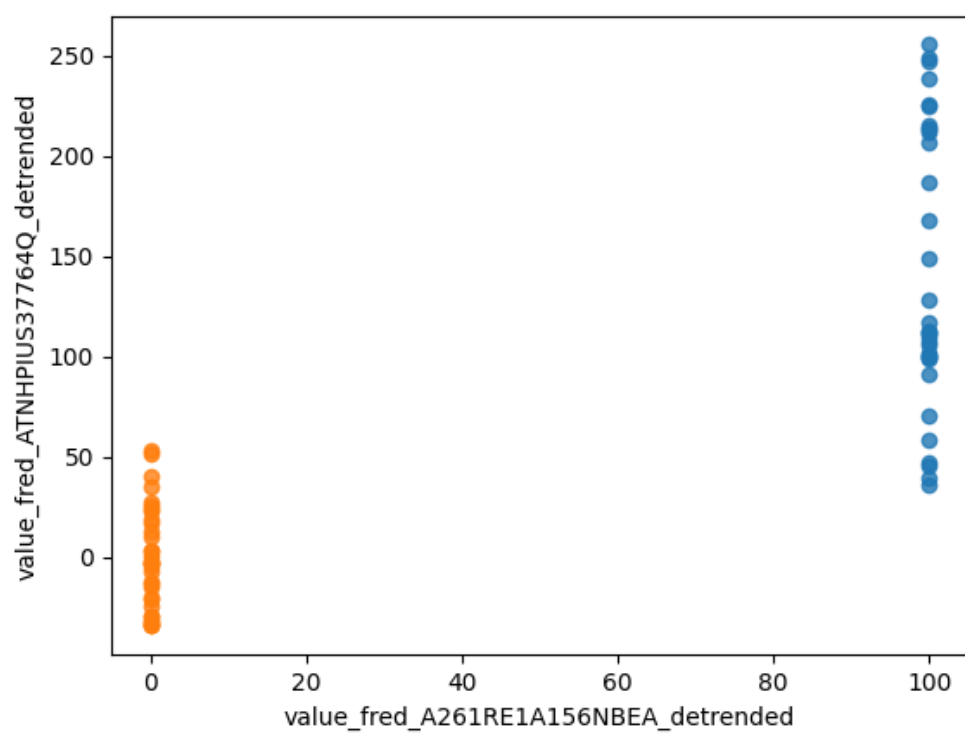


Figure 12: Detrended Regression Plot for 2026-09-26

8 Date: 2026-09-27

Series ID: CIU1010000500000I

This series is titled Employment Cost Index: Total compensation for All Civilian workers in Production, transportation, and material moving and has a frequency of Quarterly. The units are Index Dec 2005=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2001-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: USINTDCSJPY

This series is titled U.S. Intervention: With-Customer Transactions in the JPY/USD (Millions of USD) and has a frequency of Daily, 7-Day. The units are Millions of USD and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1973-03-02 and the observation end date is 2011-05-31. The popularity of this series is 4.

8.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_USINTDCSJPY	R-squared:	nan			
Model:	OLS	Adj. R-squared:	nan			
Method:	Least Squares	F-statistic:	nan			
Date:	Sun, 27 Sep 2026	Prob (F-statistic):	nan			
Time:	15:52:42	Log-Likelihood:	inf			
No. Observations:	31	AIC:	-inf			
Df Residuals:	29	BIC:	-inf			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0	0	nan	nan	0	0
value_fred_CIU1010000500000I	0	0	nan	nan	0	0
Omnibus:	nan	Durbin-Watson:	nan			
Prob(Omnibus):	nan	Jarque-Bera (JB):	nan			
Skew:	nan	Prob(JB):	nan			
Kurtosis:	nan	Cond. No.	1.18e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

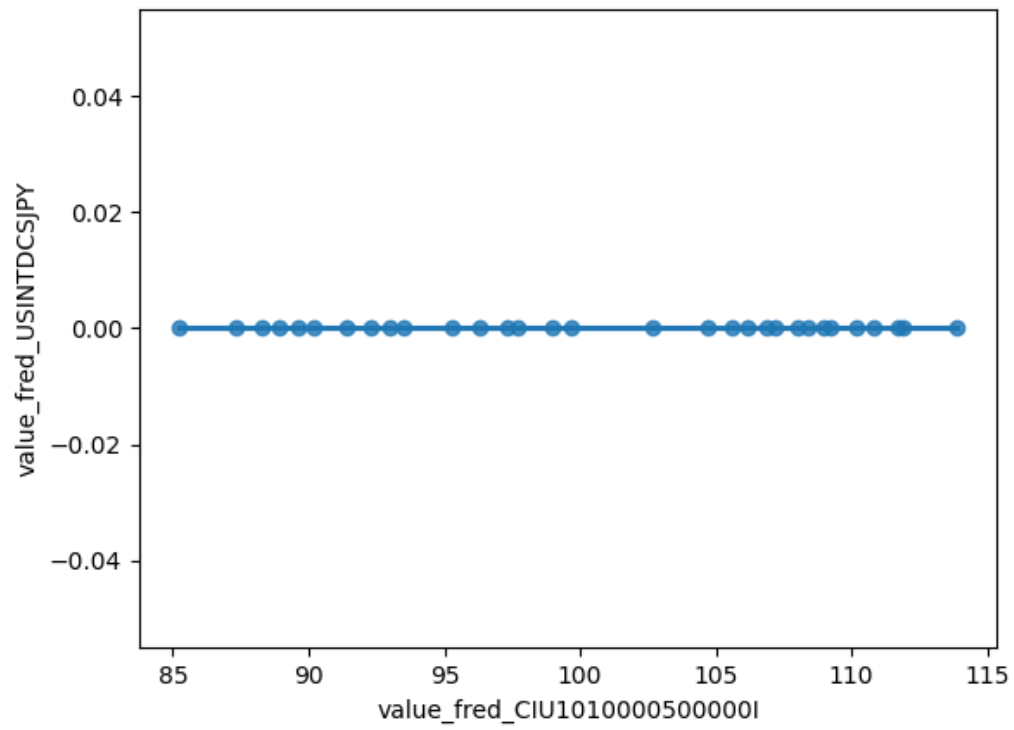


Figure 13: Raw Regression Plot for 2026-09-27

[2] The condition number is large, $1.18\text{e}+03$. This might indicate that there are strong multicollinearity or other numerical problems.

8.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_USINTDCSJPY_detrended	R-squared:	nan
Model:	OLS	Adj. R-squared:	nan
Method:	Least Squares	F-statistic:	nan
Date:	Sun, 27 Sep 2026	Prob (F-statistic):	nan
Time:	15:52:43	Log-Likelihood:	inf
No. Observations:	31	AIC:	-inf
Df Residuals:	29	BIC:	-inf
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0	0	nan	nan	0	0
value_fred_CIU1010000500000I_detrended	0	0	nan	nan	0	0

Omnibus:	nan	Durbin-Watson:	nan
Prob(Omnibus):	nan	Jarque-Bera (JB):	nan
Skew:	nan	Prob(JB):	nan
Kurtosis:	nan	Cond. No.	1.38

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

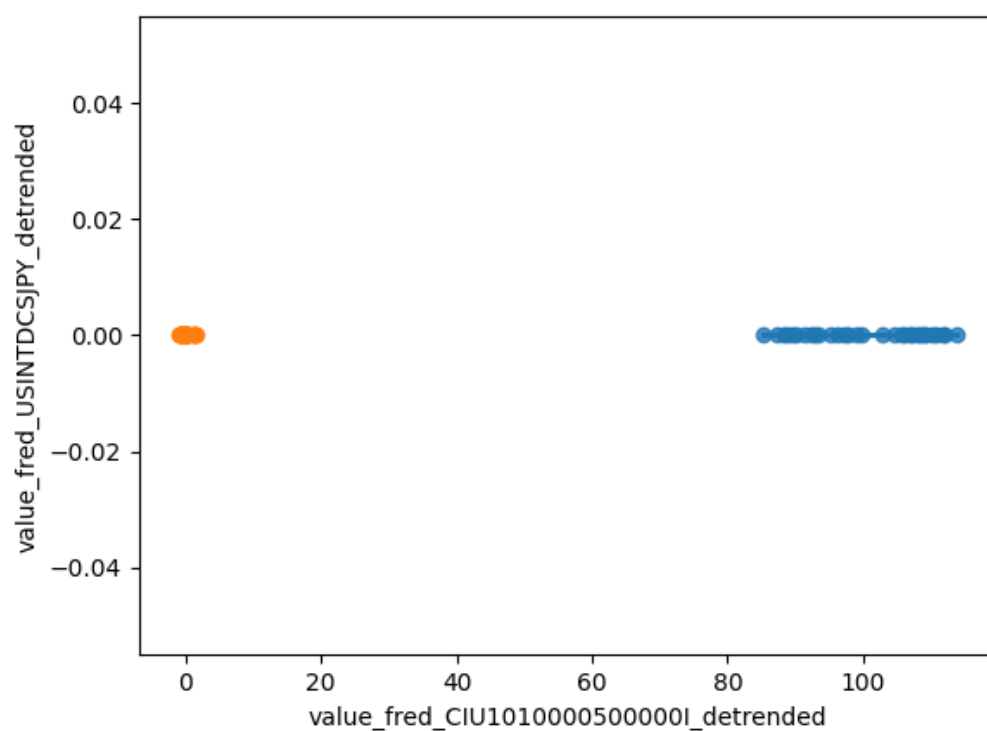


Figure 14: Detrended Regression Plot for 2026-09-27